

Technical Note: Mathematical Derivation Process of the Refractive Index Constant α

In the WRM model, the spacetime refractive index constant α is determined by the physical geometric ratio between the horizon of the observable universe (Hubble radius) and the outer boundary that defines the wave structure of the universe (1 trillion light-years).

1. Definition of the cosmic scale

- **R_obs (Hubble radius)** : Approximately 46 billion light-years (the edge of the currently observable universe).
- **R_ext (external wave boundary)** : 1 trillion light-years (the source boundary of standing waves in the WRM model).

2. Calculation of Boundary Ratio

The influence of external wave potentials on our local spacetime can be derived as a dimensionless number, expressed as the ratio of these two scales, as follows:

$$\text{Ratio} = R_{\text{obs}} / R_{\text{ext}}$$

$$\text{Ratio} = 46,000,000,000 \text{ LY} / 1,000,000,000,000 \text{ LY}$$

$$\text{Ratio} = 0.046$$

This value of **0.046 (4.6%)** represents the "modulation amplitude" of the external wave relative to the spacetime background metric, that is, the physical reality of the refractive index constant α .

3. Physical significance of the refractive index constant α

This constant α uniformly governs the "deviation from the Standard Model" in all of the following phenomena.

- **Perihelion shift fraction**: 4.6% correction to the GR theoretical value in DE440 residuals.
- **Quantum phase drift**: Phase shift of the wave function $\exp(i * \alpha * \theta)$ in the double-slit experiment.
- **Hubble anisotropy**: A 4.6% bias in the dipole axis direction in terms of expansion velocity.

Conclusion: 0.046 is inevitable.

α is not a tuning parameter (fitting variable). It is a "fundamental constant of the universe" geometrically required by the structural boundary conditions of the universe. By introducing this α , macroscopic celestial mechanics and microscopic quantum systems are completely integrated under a single scale ratio.